

1 Introduction

This document outlines 15 test cases designed to validate the core functionalities of the Caregiver Assistance System. The system includes object detection, speech synthesis, voice command processing, obstacle detection, crowd density analysis, user behavior analysis, navigation, AR overlays, and emergency detection. Each test case includes a description, preconditions, test steps, expected results, and pass/fail criteria.

2 Test Cases

ID	Test Case Description	Preconditions	Test Steps	Expected Results	Pass/Fail
1	Test object detection for identifying a person	Camera active, model file exists, person in frame	1. Start video feed. 2. Place a person within 5 meters. 3. Process frame. 4. Check detection logs.	System detects "person" with score > 0.5 and distance $< 5m$. Log entry created.	Pass if person detected and logged correctly.
2	Test speech synthesis for obstacle announcement	Speech engine initialized, obstacle detected	1. Simulate speed breaker detection. 2. Trigger announcement. 3. Monitor audio output.	System announces "Speed breaker detected at X meters." Audio is clear.	Pass if announcement is made and audible.
3	Test voice command for navigation	Microphone active, speech recognizer initialized	1. Say "Navigate to Central Park." 2. Verify destination update. 3. Check navigation steps.	Destination set to "Central Park," navigation steps provided, distance calculated.	Pass if destination and steps are logged correctly.
4	Test speed breaker detection	Camera active, speed breaker in frame	1. Start video feed. 2. Present speed breaker image. 3. Check obstacle detection.	System detects speed breaker and logs "Speed breaker at X meters."	Pass if speed breaker is detected and logged.
5	Test crowd density analysis in high-density scenario	Camera active, multiple people in frame	1. Simulate 10+ people in frame. 2. Process crowd density. 3. Check announcements.	System announces "Crowded area with 10+ people detected." Density classified as "High."	Pass if density is "High" and announcement made.

ID	Test Case Description	Preconditions	Test Steps	Expected Results	Pass/Fail
6	Test user behavior analysis for movement	Pose detection active, user in frame	1. User moves forward. 2. Process pose landmarks. 3. Check behavior logs.	System logs "Moving forward" with speed > 5.	Pass if movement is detected and logged.
7	Test OCR for text detection	Camera active, text in frame	1. Present text "STOP" in frame. 2. Wait for OCR interval. 3. Check text detection.	System detects and logs "Text detected: STOP."	Pass if text is detected and logged.
8	Test emergency detection for close object	Camera active, object in frame	1. Place car within 2 meters. 2. Process frame. 3. Check alerts.	System logs "Warning: car detected at X meters" and sets emergency flag.	Pass if alert is logged and flag set.
9	Test navigation AR overlay	Navigation active, road context detected	1. Set destination. 2. Simulate road context. 3. Check frame for AR arrow.	Arrow overlay appears at (25, 25) in frame.	Pass if arrow is overlaid correctly.
10	Test caregiver dashboard log updates	Log file exists, dashboard active	1. Generate log entries. 2. Access caregiver dashboard. 3. Check log updates.	Dashboard displays last 20 logs with correct classes (e.g., danger, info).	Pass if logs are displayed correctly.
11	Test voice command for caregiver phone update	Microphone active, speech recognizer initialized	1. Say "Update caregiver phone." 2. Say "123-456-7890." 3. Check user data.	Caregiver phone updated to "123-456-7890" and logged.	Pass if phone is updated and logged.
12	Test optical flow for moving object detection	Camera active, moving object in frame	1. Simulate moving car. 2. Process optical flow. 3. Check moving objects.	System detects "car approaching from the right" and logs.	Pass if moving object is detected and logged.
13	Test user gesture detection (waving)	Pose detection active, user in frame	1. User waves left hand. 2. Process pose landmarks. 3. Check behavior logs.	System logs "Waving left hand" in behavior.	Pass if gesture is detected and logged.

ID	Test Case Description	Preconditions	Test Steps	Expected Results	Pass/Fail
14	Test road context detection	Camera active, road lines in frame	1. Present road with lane lines. 2. Process frame. 3. Check road context.	System detects road context and logs lane detection.	Pass if road context is detected.
15	Test logout functionality	System active, user logged in	1. Say "Logout." 2. Check WebSocket event. 3. Verify log entry.	System emits 'logout' event and logs "User logged out."	Pass if logout event is emitted and logged.

Table 1: Test Cases for Caregiver Assistance System

3 Conclusion

These test cases cover the core functionalities of the Caregiver Assistance System, ensuring robust performance in object detection, speech interaction, navigation, and safety features. Each test case is designed to validate specific system behavior under controlled conditions.